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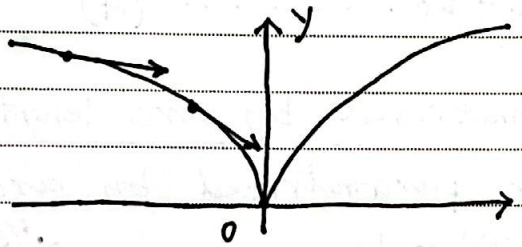
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- Course title: Differential geometry for curves and surfaces.
- Course code: MAT 313
- Text book: Differential geometry of curves and surfaces by Manfredo P. do Carmo

Back ground discussion

A parametrized differentiable curve is a differentiable map $\alpha: I \to \mathbb{R}^3$ of an open interval $I = (a,b)$ of $\mathbb{R}$ into $\mathbb{R}^3$.

Example 2 The map $\alpha: \mathbb{R} \to \mathbb{R}^3$ by $\alpha(t) = (t^3, t^2)$, $t \in \mathbb{R}$ is a parametrized differentiable curve which has the trace



here $x = t^3$
$y = t^2$
Eliminate t to obtain,
$y^3 = x^2$

In this example, $\alpha'(t) = (3t^2, 2t)$

At, $t = t_0$, the slope of the tangent line is $m = \frac{2t_0}{3t_0^2} = \frac{2}{3t_0}$

And the tangent line goes Through $\alpha(t_0) = (t_0^3, t_0^2)$

Hence, the equation of the tangent at $t = t_0$ is

$$y - t_0^2 = \frac{2}{3t_0}(x - t_0^3)$$

Definition Let $\alpha: I \to \mathbb{R}^3$ be a parametrized differentiable curve. For each $t \in I$ where $\alpha'(t) \neq 0$, there is a well-defined straight line which goes through $\alpha(t)$ and contains the vector $\alpha'(t)$. This line is called the tangent line to α at t. Any point t where $\alpha'(t) = 0$ is a singular point of α. We restrict our attention to curves without any singular points. In example 2, $t = 0$ is a singular point.

- A parametrized differentiable curve $\alpha: I \to \mathbb{R}^3$ is said to be regular if $\alpha'(t) \neq 0$ $\forall\, t \in I$.

Given $t \in I$, the arc length of a regular parametrized curve $\alpha: I \to \mathbb{R}^3$, from the pt. t_0 is $s(t) = \int_{t_0}^{t} |\alpha'(t)|\, dt$, where ... (1)

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$$|\alpha'(t)| = \sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2} \quad \ldots\ldots\ldots (2)$$

$$\alpha(t) = (x(t), y(t), z(t)), \quad \alpha'(t) = (x'(t), y'(t), z'(t)).$$

The arc length $s(t)$ is a differentiable function of t

$$s(t) = \int_{t_0}^{t} |\alpha'(u)|\, du$$

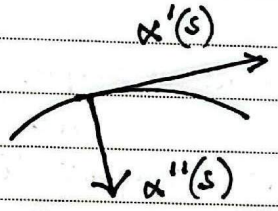
$$\Rightarrow s'(t) = |\alpha'(t)| \quad \ldots\ldots\ldots (3)$$

It may well be the case that the parameter t itself is the arc length s, i.e., $s = t$. Then $\frac{ds}{dt} = 1 = |\alpha'(t)|$

And $$s = \int_{t_0}^{t} |\alpha'(u)|\, du = t - t_0 \quad \ldots\ldots\ldots (4)$$

The local theory of curves parametrized by arc length.

Let $\alpha: I = (a, b) \to \mathbb{R}^3$ be a curve parametrized by arc length s. We know that the speed is $|\alpha'(s)| = 1$. Hence, the acceleration is normal to the tangent vector at the point. This acceleration vector $\alpha''(s)$ pulls the object and deviates it from the tangent line. If it were zero, the object would have travelled along the tangent line. The more the acceleration, i.e., the more $|\alpha''(s)|$ is, the more the deviation of the trajectory of the object or curve from the tangent line at s.



Definition Let $\alpha: I \to \mathbb{R}^3$ be a curve parametrized by arc length $s \in I$. The number $|\alpha''(s)| = k(s)$ is called the curvature of α at s.

Orientation a la do Carmo: The ambient space where the curves and surfaces are living in is $\mathbb{R}^3$. The curve $\alpha(s)$ living in $\mathbb{R}^3$ has a velocity vector

$$\alpha'(s) = \begin{bmatrix} x'(s) \\ y'(s) \\ z'(s) \end{bmatrix} \quad \ldots\ldots\ldots (5)$$

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To discuss the orientation of this curve reversed the direction of traversal as the parameter of the curve increases do Carmov

Recall the curvature $k(s)$ of α at s, given by $k(s) = |\alpha''(s)|$ ..... (6)

At points where $k(s) \neq 0$, a unit vector $n(s)$ in the direction of $\alpha'(s)$ is well-defined by the equation $\alpha''(s) = k(s)n(s)$ ..... (7)

From $\alpha'(s) \cdot \alpha'(s) = 1$, one obtains by differentiation $\alpha''(s) \cdot \alpha'(s) = 0$ and from (7) one finds that $n(s)$ is normal to $\alpha'(s)$ and is called the normal vector at s.

The plane determined by the unit tangent vector $\alpha'(s)$ and the normal vector $n(s)$, is called the osculating plane.

- Singular points of order 0 and 1

For the arc-parametrized curve $\alpha(s)$, s is a singular point of order 0 if $\alpha'(s) = 0$ and it is a singular point of order 1 if $\alpha''(s) = 0$ with $\alpha'(s) \neq 0$

Example: (planar inflection curve) $\alpha(t) = (t, t^3, 0)$ is an inflection curve lying in the x-y plane. Note that it's not arc-parametrized. In fact, to understand singular pts of order 0 and 1, one doesn't need an arc-par-ametrized curve necessarily. In this example, $\alpha'(t) = (1, 3t^2, 0)$ reflecting the fact that $\alpha'(t) \neq 0$. It means that the curve has no singular points of order 0. It is a regular curve. One also sees that $\alpha''(t) = (0, 6t, 0)$ so that $\alpha''(0) = (0,0,0)$ i.e., 0 is a singular point of order 1.

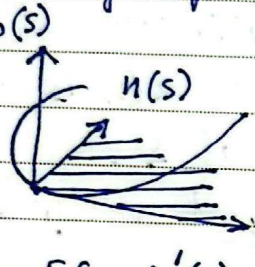
In what follows, we restrict ourselves to arc-parametrized curve (which is automatically regular) with no singular point of order 1.

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We shall denote by $t(s) = \alpha'(s)$ the unit tangent vector of α at s so that $t'(s) = \alpha''(s) = k(s)\,n(s)$

$$\boxed{\begin{aligned} t(s) &= \alpha'(s) \\ t'(s) &= \alpha''(s) = k(s)\,n(s) \end{aligned}} \quad \cdots\cdots\cdots (8)$$

The unit vector $t(s) \wedge n(s) = b(s)$ is normal to the osculating plane and is called <u>binormal vector</u> at s.



$$t(s) \wedge n(s) = b(s) \quad \cdots\cdots\cdots (9)$$

$b'(s)$ measures the extent with which the curve is pulled out of the osculating plane. If $b'(s) = 0$, the curve always stays on the osculating plane and is never pulled out off it.

From $b(s) = t(s) \wedge n(s)$, one has

$$b'(s) = t'(s) \wedge n(s) + t(s) \wedge n'(s) = t(s) \wedge n'(s) \quad \cdots\cdots\cdots (10)$$

; [$\because$ $t'(s)$ and $n(s)$ are collinear]

In other words, $b'(s) \perp t(s)$. And $b'(s)$ is perpendicular to $b(s)$ itself. And the only way they are satisfied is $b'(s)$ is parallel to $n(s)$. i.e., $b'(s) \parallel n(s)$ $\cdots\cdots\cdots (11)$

And it prompts us to write $\boxed{b'(s) = \tau(s)\, n(s)}$ $\cdots\cdots (12)$ for some function $\tau(s)$.

<u>Definition (Torsion)</u>

Let $\alpha : I \to \mathbb{R}^3$ be a curve parametrized by arc length s such that $\alpha''(s) \neq 0$, $s \in I$. The number $\tau(s)$ defined by $b'(s) = \tau(s)\, n(s)$ is called the <u>torsion</u> of α at s.

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Remark.

If α is a plane curve (i.e., $\alpha(I)$ is contained in a plane), then the plane of the curve agrees with the osculating plane; hence $\tau \equiv 0$. Conversely, if $\tau \equiv 0$ (and $k \neq 0$) we have $b'(s) = 0$, so that $b(s) = b_0 = \text{constant}$ ..... (13)

And therefore,

$$(\alpha(s) \cdot b_0)' = \alpha'(s) \cdot b_0 = 0 \; ; \; [\because b(s) \text{ is normal to } \alpha'(s) = t(s)]$$

This implies $\alpha(s) \cdot b_0 = \text{constant}$ ..... (14)

$\forall s \in I$

We have $\alpha(s) \cdot b_0 = d = \text{some constant}$ for all $s \in I$. For an arbitrary reference point $s_0 \in I$, $[\alpha(s) - \alpha(s_0)] \cdot b_0 = 0$ meaning that the vectors joining any two points on the curve $\alpha(s)$ will always be perpendicular to b_0. This means that the curve $\alpha(s)$ lies strictly in a plane perpendicular to b_0.

Note that $k \neq 0$ is essential throughout as

$$n(s) = \frac{\alpha''(s)}{k(s)} \quad \ldots\ldots\ldots (15)$$

if $k(s) = 0$ anywhere $n(s)$ ceases to exist and since $b(s) = t(s) \wedge n(s)$, the existence of the constant vector b_0 collapses.

Summary: To each value of the parameters, we have associated three orthogonal unit vectors $t(s)$, $n(s)$, $b(s)$. The trihedron thus formed is referred to as the **Frenet trihedron** at s. The derivatives $t'(s) = k(s)\, n(s)$, $b'(s) = \tau(s)\, n(s)$ of the vectors $t(s)$ and $b(s)$, when expressed in the basis $\{t, n, b\}$, yield geometric entities curvature k and torsion τ which give us information about the

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behaviour of α in the neighbourhood of s.

Let us now compute $n'(s)$ using $n(s) = b(s) \wedge t(s)$

$$n'(s) = b'(s) \wedge t(s) + b(s) \wedge t'(s)$$

$$= -\tau(s)\,(n(s) \wedge t(s)) + k(s)\,(b(s) \wedge n(s))$$

: [using $b'(s) = -\tau(s)\,n(s)$ and $t'(s) = k(s)\,n(s)$]

$$= -\tau(s)\,(-b(s)) + k(s)\,(-t(s))$$

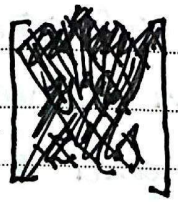
$$= -\tau(s)\,b(s) - k(s)\,t(s) \quad \ldots\ldots\ldots\ldots (16)$$

We thus have Frenet formulae

$$\boxed{\begin{aligned} t'(s) &= k(s)\,n(s) \\ n'(s) &= -\tau(s)\,b(s) - k(s)\,t(s) \\ b'(s) &= \tau(s)\,n(s) \end{aligned}} \quad \ldots\ldots\ldots\ldots (17)$$

The tb plane is called rectifying plane and the nb plane is called the normal plane. The line that contains $n(s)$ and passes through $\alpha(s)$ is called the principal normal. And the line that contains $b(s)$ and passes through $\alpha(s)$ is called the binormal. The inverse $R = \frac{1}{k}$ of the curvature is called the radius of curvature at s.

Now, let us get back to the orientation discussion of the curve $\alpha(s)$ ~~[illegible]~~.



As $-s$ increases the original parameter s decreases

Define $\beta(-s) = \alpha(s)$ ..... (18)

so that $\beta'(-s) = \frac{d\beta}{d(-s)}(-s) = \frac{d\alpha(-s)}{d(-s)} = -\frac{d\alpha(s)}{ds} = -\alpha'(s)$

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$\therefore \beta'(-s) = -\alpha'(s) \cdots\cdots (19)$

And $\beta''(-s) = \frac{d\beta'}{d(-s)}(-s) = \frac{d}{d(-s)}(-\alpha'(s)) = \frac{d}{ds}(\alpha'(s)) = \alpha''(s)$

$\therefore \beta''(-s) = \alpha''(s) \cdots\cdots (20)$

Therefore, $\alpha'(s)$ changes sign if the orientation of the curve reverses while $\alpha''(s)$ and hence curvature remains invariant under orientation reversal of the curve.

Fundamental theorem of the local theory of curves.

Given differentiable functions $k(s) > 0$ (recall $k(s) = |\alpha''(s)|$ so that $k(s) \geq 0$ and $k(s) = 0$ is not allowed as $n(s)$ is defined through $n(s) = \frac{\alpha''(s)}{k(s)}$) and $\tau(s)$, $s \in I$, there exists a regular parametrized curve $\alpha: I \to \mathbb{R}^3$ such that s is the arc length, $k(s)$ is the curvature, and $\tau(s)$ is the torsion of α. (The regular parametrized curve here is actually arc length parametrized which is automatically regular — the existence of such an arc length parametrized curve is claimed in the fundamental theorem). Moreover, any other curve $\tilde{\alpha}$ satisfying the same conditions, differs from α by a rigid motion; that is there exists an orthogonal linear map ρ of $\mathbb{R}^3$, with positive determinant, and a vector c such that $\tilde{\alpha} = \rho \circ \alpha + c$.

- In the next class, we shall prove the fundamental theorem of local theory of curves.

□